

Adaptive Experimentation under Robustness

Abhi Vemulapati

- Box – "All models are wrong, but some are useful"
- Subject to *ambiguity* and *misspecification*
 - Ambiguity – inability to commit to one of many possible models
 - Misspecification – Posited models only approximate true DGP
- Much work on "robustness" in dynamic economic models
 - Robustness – how to adjust decisions under ambiguity/misspecification
- Recent work suggests interesting applications to econometrics
 - Little yet to be said about robustness in sequential experimentation

- Why is robustness useful for studying adaptive experiments?
 - Insures against a rich class of econometric modeling concerns
- How to incorporate robustness into decision making?
 - Apply setup of Cerreia-Vioglio et al. ('26)
- How to design asymptotically optimal decisions under robustness?
 - For one class of problems: Construct robust PDEs
 - For other class: Solve minimax games in the limit

Why is robustness useful?

Example 1: Dynamic Treatment Allocation

- Common issue with dynamic treatment allocation is *distribution shifts* or *surrogate outcomes*...
- ...i.e. when treatment effect being estimated is not reflective of population
 - Example: Community efforts drive high trial participation in some areas but not others
- Result: For treatment estimation/assignment, Neyman allocation remains optimal
 - Also, employ efficient estimators
 - With optimal stopping, misspecification changes horizon

Example 2: Multi-Armed Bandits

- Consider: K -armed bandit experiment
- Arm k reward $\sim N(\mu_k, \sigma_k^2)$ i.i.d. across time, arms
- Large literature on bandits with non-iid rewards, but no decision-theoretic underpinnings
- Robustness: Microfound nonstationarity in bandits
 - rewards no longer normal, i.i.d. across time, possibly shifted w.r.t. μ_k
- Result: Optimal policy still deterministic
 - Due to misspecification, value learning less

Example 3: Forecasting

- Consider state-space model

$$y_t = A\theta_t + \varepsilon_t; \quad \theta_t = B\theta_{t-1} + \nu_t.$$
$$(\varepsilon_t, \nu_t) \sim N(0, \Sigma).$$

- Robustness: ε_t, ν_t misspecified (non-Gaussian, nonzero mean)
 - Result: use biased estimate to forecast y_t
- Or: ambiguity over structural components of A, B and misspecification of $y_t, \theta_t | A, B$

- Misspecification: Hansen and Sargent ('06, '07, '11), Cerreia-Vioglio ('26)
 - **This paper:** Link discrete time and continuous time formulations of robust decision problems
 - Apply framework of Cerreia-Vioglio ('26)
- Local asymptotics: Hirano and Porter ('23), Adusumilli ('26)
 - Tools for studying asymptotic behavior
- Ambiguity and misspecification for econometrics: Adusumilli ('26), Andrews et al. ('26)
 - **This paper:** Extends ideas from both to dynamic setting
 - However, we relax prior ambiguity in some cases

Models, ambiguity, and misspecification

Defining Robustness

- We follow Cerreia-Vioglio et al. ('26)
 - Provides method to define plausible models
- First, introduce set $\mathcal{Q}$ of structured (ambiguous) models
- Then, surround each model in $\mathcal{Q}$ w/ misspecified alternatives
- How we define robustness depends on definition of model
 - Based on def'ns, categorize Examples 1-3 into two classes of robustness
 - However, resulting decision problems will be the same

Decision theory

- Over n periods, statistician sequentially observes i.i.d. data $Y_{1:n} \in \mathcal{Y}$
- There is some unknown parameter
 - Will usually refer to as $\theta \in \Theta$ or $\gamma \in \Gamma$
- In period $i \in [n]$:
 - 1 Take action A_i
 - 2 Draw Y_i
 - 3 Observe utility $u(\cdot, A_i)$
- Total (realized) utility:

$$U_n(\cdot, A_{1:n}) := \sum_{i=1}^n u(\cdot, A_i)$$

- $u(\cdot, A_i)$ takes Y_i and/or θ as arguments

- Micro theory: **model** $\equiv$ joint distribution over payoff-relevant states
 - Payoff-relevant states based on inputs of utility function
- When utility does not contain parameters, call this a γ -*model*:

$$m(Y_{1:n}|\gamma) := \prod_{i=1}^n m(Y_i|\gamma)$$

- When utility contains θ , use a *Bayesian model*:

$$m(Y_{1:n}, \theta) := m_0(\theta) \otimes m(Y_{1:n}|\theta)$$

- Bayesian model $\equiv$ likelihood paired with prior m_0
- m_0 encodes lack of knowledge over θ
- l'hood links data to θ

Structured models

- Structured models: Set of possible "baseline" models

- For γ -models:

$$\mathcal{Q}^{\Gamma} := \{p(Y_{1:n}|\gamma) : \gamma \in \Gamma\}$$

- For Bayesian models (Adusumilli '26):

$$\mathcal{Q}^{\Delta(\Theta)} := \{p_0(\theta) \otimes p(Y_{1:n}|\theta) : p_0(\cdot) \in \Delta(\Theta)\}$$

- Ambiguity: Inability to commit to a single *model*
 - Adusumilli ('26): ambiguity over *models* $\equiv$ ambiguity over *priors*

Unstructured models and misspecification

- Surround each structured model with misspecified alternatives

$$\mathcal{M} := \left\{ m \in \Delta(\mathcal{S}) : \min_{q \in \mathcal{Q}} R_q(m) \leq K \right\}$$

- $R_q(m) = \text{KL}(m||q)$ is relative entropy
- $\mathcal{S}$ is support of payoff-relevant states
 - For γ -models: $\mathcal{S} = \mathcal{Y}$
 - For Bayesian models: $\mathcal{S} = \mathcal{Y} \times \Theta$

A Unified Example

- Suppose we have data $Y_{1:n}$ on vaccine outcomes
 - Want to administer vaccine globally
- If utility doesn't contain parameters (e.g., want to predict Y_i):
 - Ambiguity is over *best approximating* γ -model (e.g., $Y_i|\gamma \sim N(\gamma, 1)$)
 - Misspecification: $Y_{1:n}|\gamma$ possibly shifted, no longer i.i.d.
- Now, if we want to estimate ATE θ (i.e., enters utility)
 - Ambiguity: Cannot form a prior over θ
 - Another type of misspecification: trial only took place in US
 - $\Rightarrow$ we are observing $Y|\theta^{\text{US}}$ instead of $Y|\theta$!
 - So, link between data & structural θ misspecified

Robust decision making

Variational preferences

- Follows from Cerreia-Vioglio et al. ('26)
- Maximize utility under worst-case model $m \in \mathcal{M}$:

$$\max_{\{A_i\}_i} \min_{m \in \mathcal{M}} \left\{ \mathbb{E}_m [U_n(\cdot, A_{1:n})] : \min_{q \in \mathcal{Q}} R_q(m) \leq K \right\}$$

- Optimal value is attained at

$$\max_{\{A_i\}_i} \min_{q \in \mathcal{Q}} -\lambda^{-1} \log \mathbb{E}_q \left[\exp \left\{ -\frac{U_n(\cdot)}{\lambda} \right\} \right]$$

- $\lambda \Leftrightarrow K$ is Lagrange multiplier

Decision Problem

- How to interpret $\min_{q \in \mathcal{Q}}$?
- For γ -models: Optimal action satisfies

$$A^* = \arg \min_{\{A_i\}_i} \max_{\gamma \in \Gamma} \mathbb{E}_{\gamma} \left[\exp \left\{ -\frac{U_n(Y_{1:n}, A_{1:n})}{\lambda} \right\} \right]$$

- For Bayesian models (Adusumilli '26):

$$A^* = \arg \min_{\{A_i\}_i} \max_{p_0 \in \Delta(\Theta)} \mathbb{E}_{p_0} \left[\exp \left\{ -\frac{U_n(\theta, Y_{1:n}, A_{1:n})}{\lambda} \right\} \right]$$

- Effectively the same problem, though interpretation changes

Local asymptotics and diffusions

Local Asymptotics

- Local asymptotics introduces "weak signal" scaling:

$$t := \frac{j}{n}, \quad Y_{j,n} := \frac{Y_j}{\sqrt{n}}, \quad \theta_n = \frac{\theta}{\sqrt{n}}.$$

- t : Fraction of the experiment that has passed
 - Equivalently, $j = \lfloor nt \rfloor$
 - Idea: Even if observations not Gaussian in discrete time, treat them as such in limit
 - Scaling ensures problem nontrivial in limit
- Scaling utility for continuous action spaces

$$u_n(Y_{j,n}, A_{j,n}) \equiv u(\sqrt{n}Y_{j,n}, \sqrt{n}A_{j,n})$$

- Under quadratic approximation, can work with $u(Y_{j,n}, A_{j,n})$
- For sampling problems, A_i does not need scaling

- For large n , only $(\{x_a(t), q_a(t)\}_{a=0}^1, t)$ needed to update beliefs

$$x_a(t) := \frac{1}{\sqrt{n}} \sum_{i=1}^{\lfloor nt \rfloor} Y_i^{(a)} \mathbb{I}[A_i = a]; \quad q_a(t) := \frac{1}{n} \sum_{i=1}^{\lfloor nt \rfloor} \mathbb{I}[A_i = a]$$

- $x_a(t)$: Cumulative outcomes from sampling treatment a up to time t
- $q_a(t)$: Time spent observing a up to time t

Bayesian models in the limit

Bayesian Models in the Limit

- Despite difficulty of dynamic approaches, can experiment under commitment
 - Not an unreasonable constraint
 - e.g., drug companies often required to commit to pre-specified trial designs
- Focus on adaptive sampling with best-arm identification (BAI) or estimation
 - Treatments are assigned sequentially using (possibly) randomized rule
 - Sample until stopping rule is satisfied
 - At end of experiment, employ an estimator δ

- Two treatments $a \in \{0, 1\}$
 - Generate signals $Y_i^{(a)} \sim N(\mu_a, \sigma_a^2)$, σ_a^2 known
- Parameter of interest: $\theta := (\mu_1, \mu_0)$
 - Interpret as treatment effects
- Consider continuous limit problem directly
 - Experiment runs for time $t \in [0, 1]$
- DM samples according to $\{\pi_0(t), \pi_1(t)\}$
 - $\pi_a(t)$: Probability of observing treatment a at period t

- Recall local asymptotic scaling
- Under baseline l'hood, describe sufficient states in limit:

$$\begin{aligned}dx_a(t) &= \pi_a(t)\mu_a dt + \sqrt{\pi_a(t)}\sigma_a dW_a(t); \\dq_a(t) &= \pi_a(t)dt\end{aligned}$$

- Sample with $\pi = (\pi_1(t), \pi_0(t))$ until stopping time τ
- At time τ : Choose terminal action δ_τ
- Estimation: δ_τ targets $\mu_1 - \mu_0$
- BAI: $\delta_\tau \in \{0, 1\}$ chooses arm with higher posterior mean

Decision Problems

- Let $\rho(\cdot)$ be bowl-shaped loss function
 - i.e., $\{x : \rho(x) \leq c\}$ convex and symmetric for all x
- Estimation: Recall variational form

$$\min_{\pi, \delta, \tau} \max_{p_0 \in \Delta(\Theta)} \mathbb{E} \left[\exp \left\{ \lambda^{-1} (\rho(\delta_\tau - (\mu_1 - \mu_0)) + c\tau) \right\} \right].$$

- Similarly, for BAI: evaluate decision under tilted regret

$$\min_{\pi, \delta, \tau} \max_{p_0 \in \Delta(\Theta)} \mathbb{E} \left[\exp \left\{ \lambda^{-1} (\max\{\mu_1 - \mu_0, 0\} - (\mu_1 - \mu_0)\delta_\tau + c\tau) \right\} \right]$$

- Sharp result: Neyman allocation remains optimal

$$(\pi_1^*(t), \pi_0^*(t)) := \left(\frac{\sigma_1}{\sigma_1 + \sigma_0}, \frac{\sigma_0}{\sigma_1 + \sigma_0} \right).$$

- At any t : Consider efficient estimator

$$m_t = \frac{\sigma_1 + \sigma_0}{t} (\sigma_1^{-1} x_1(t) - \sigma_0^{-1} x_0(t))$$

- Choose $\delta_\tau^* = m_\tau$ for estimation and

$$\delta_\tau^{BAI} = \mathbb{I}[m_\tau \geq 0],$$

for treatment assignment

Optimal Stopping

- For estimation, stopping rule is deterministic

$$\tau^* := \min_t \mathbb{E} \left[\exp \left\{ \lambda^{-1} \left(\rho \left(t^{-1/2} Z \right) + ct \right) \right\} \right]$$

- where $Z \sim N(0, (\sigma_1 + \sigma_0)^2)$
- For BAI, sample until tm_t exits a symmetric boundary:

$$\tau_{BAI}^* := \inf_t \{t | m_t| \geq b\}$$

- Unlike (π, δ) , stopping boundaries depend on λ

Application: Wald's Problem

- Wald (1951) considers drift estimation under squared loss $\rho(x) = x^2$ and optimal stopping
- Optimal stopping time grows with misspecification ($\lambda \rightarrow 0$):

$$\tau^* = \frac{\sigma^2}{\lambda} + \frac{\sigma}{\sqrt{c}} \sqrt{1 + \frac{c\sigma^2}{\lambda^2}}$$

- where $\sigma := \sigma_1 + \sigma_0$
- Employ Neyman allocation up to τ^* and then use $\delta_{\tau^*}^*$

- Consider the following types of priors for each problem
 - BAI: symmetric prior w/ support $\pm(\sigma_1\Delta, -\sigma_0\Delta)$ for some Δ
 - Estimation: Improper Gaussian prior $N(0, \infty)$
- Under these priors: indifference between all sampling rules!
 - As noted in Adusumilli ('25), Armstrong ('26)
- Then, optimal to employ Neyman allocation
 - As long as loss is symmetric in ATE (i.e., $\mu_1 - \mu_0$)
 - Misspecification is symmetric $\Rightarrow$ still applies
- Neyman allocation minimizes posterior variance of $\mu_1 - \mu_0$
 - Independent of specification of $\rho(\cdot)$
 - So, works under exponential tilting

Discussion (cont.)

- Symmetricity also implies MLE is optimal
 - Since $\exp\{\rho(\cdot)/\lambda\}$ is still bowl-shaped
 - Goes back to Blyth ('51)
- Stopping time then solved for independently
- Interestingly, cannot say if misspecification causes more sampling for estimation problems
 - May use fewer samples for heavily bounded losses (e.g., $\mathbb{1}[|x| \geq 1]$)

γ -models in the limit

- In Example 1, considered limit problem directly
- For γ -models, recursive solution often needed
- Consider discrete estimation of SDEs $x(t), q(t)$
- Can then construct PDE limits under robustness
- First, look at general action spaces over A_i
 - Generalizes recursive estimation and sampling problems
 - Ignore q for now

General Setup – γ -Models

- Assume rewards are drawn from $Y_i | \gamma \stackrel{iid}{\sim} N(\gamma, \sigma^2)$
- In many problems, minimizing over γ leads to trivial risk
- Instead, consider *smooth ambiguity* – prior p_Γ over γ -models
- Also need $\phi_\Gamma(\cdot)$, which governs aversion to ambiguity

$$\max_{\{A_i\}} \int_{\Gamma} \phi_{\Gamma} \left(-\lambda \log \mathbb{E}_{Y_{1:n}} \left[\exp \left(-\frac{1}{\lambda} \sum_{i=1}^n u(Y_i, A_i) \right) \right] \right) dp_{\Gamma}(\gamma)$$

- Setting $\phi_t(\cdot) = -\exp\{-t/\xi\}$, we get

$$\max_{\{A_i\}} \int_{\Gamma} \left(\mathbb{E}_{Y_{1:n}} \left[\exp \left(-\frac{1}{\lambda} \sum_{i=1}^n u(Y_i, A_i) \right) \right] \right)^{-\lambda/\xi} dp_{\Gamma}(\gamma)$$

- ξ is aversion to ambiguity, λ is sensitivity to misspecification
- For tractability, take these to be same ($\lambda = \xi$)
- Then, problem reduces to simple prior with exponentially tilted utility

Variational Form for Bandits

- Define information up to period j :

$$s_j := \{Y_1, \dots, Y_{j-1}\}$$

- In recursive form, $V(s_j)$ is remaining value in state s_j :

$$V(s_j) = \max_{A_j} -\lambda \log \mathbb{E}_{Y_{j:n}|s_j} \left[\exp \left(-\frac{1}{\lambda} (u(Y_j, A_j) + V(s_{j+1})) \right) \right]$$

- Local asymptotics reduces dimension of s_j

State Transition

- Write $Y_{t,n}$ as $\gamma/n + \varepsilon_t/\sqrt{n}$ where $\varepsilon_t \sim N(0, \sigma^2)$
- State transition:

$$s_{t+1/n} = \left(x(t) + \left(\frac{\gamma}{n} + \frac{\varepsilon_t}{\sqrt{n}} \right), t + \frac{1}{n} \right)$$

- Recursive form under weak signal scaling:

$$V(s_t) = \max_A -\lambda \log \mathbb{E}_{|s_t} \left[\exp \left(-\frac{1}{\lambda} (u(Y_{t,n}, A_{t,n}) + V(s_{t+1/n})) \right) \right]$$

- Now, employ three approximations to get an HJB

- First approximation uses Ito's Lemma: $V(s_{t+1/n}) - V(s_t) \approx$

$$\frac{1}{n} V_t + \left(\frac{\gamma}{n} + \frac{\varepsilon_t}{\sqrt{n}} \right) V_x + \frac{\sigma^2}{2n} V_{xx}$$

- Define

$$R_{t,n} := u(Y_{t,n}, A_{t,n}) + V(s_{t+1/n}) - V(s_t)$$

- Recursive form simplifies to

$$0 = \max_A -\lambda \log \mathbb{E}_{\cdot|s_t} \left[\exp \left(-\frac{1}{\lambda} R_t \right) \right].$$

CGFs and Mean-Variance Optimization

- Cumulant generating function provide a novel representation

$$-\lambda \log \mathbb{E}_{\cdot|s_t} \left[\exp \left(-\frac{1}{\lambda} R_t \right) \right] \approx \mathbb{E}_{\cdot|s_t} [R_t] - \frac{1}{\lambda} \mathbb{V}_{\cdot|s_t} [R_t]$$

- Typical criticisms of mean-variance optimization
 - Violates axioms of EU representation
 - Only valid under Gaussian lotteries
- We extend its microfoundation to dynamic settings
 - In local asymptotic limit, robustness $\Leftrightarrow$ mean-variance optimization

Local Utility Approximation

- Third approximation localizes $u(Y, A)$ around $Y = 0$:

$$u(Y, A) \approx u(0, A) + u_Y(0, A)Y + \frac{1}{2}u_{YY}(0, A)Y^2$$

- Let $\gamma(s) := \mathbb{E}_{\cdot|s}[\gamma]$. Taking $n \rightarrow \infty$:

$$\begin{aligned} 0 = \max_A V_t + u(0, A) + u_Y(0, A)\gamma(s) + \frac{\sigma^2}{2}u_{YY}(0, A) \\ + \gamma(s)V_x + \frac{\sigma^2}{2}V_{xx} - \frac{\sigma^2}{2\lambda}(u_Y(0, A) + V_x)^2 \end{aligned}$$

- Optimal action balances usual optimization with *risk-sensitivity*
 - Hence the quadratic penalty term
- Risk-sensitive penalty captures two channels of misspecification:
 - Distortion of true random component of rewards
 - Distortion of link between Y and parameter
- $u_Y(0, A)$ captures the former channel

Application to Bandit Problems

- Consider one-armed bandit: risky arm 1 with $Y_j \sim N(\gamma, \sigma^2)$, safe arm 0 with reward 0
- At each t : choose randomization policy $\pi_t \in [0, 1]$ over arms $A_t \in \{0, 1\}$
 - π is probability of pulling arm 1
- Welfare utility: $u(Y, A) = AY$
- Reintroduce additional state variable q

- Apply the central result and maximize over π :

$$0 = V_t + \max \left\{ \gamma(s) + \gamma(s) V_x + V_q + \frac{\sigma^2}{2} V_{xx} - (1 + V_x)^2, 0 \right\}$$

- Without misspecification (Adusumilli '25):

$$0 = V_t + \max \left\{ \gamma(s) + \gamma(s) V_x + V_q + \frac{\sigma^2}{2} V_{xx}, 0 \right\}$$

- Robust policy is index-based ($\pi_t \in \{0, 1\}$ for all t)
 - In particular, if $\pi_t = 0$ at any point, never pull again

Optimal Policy

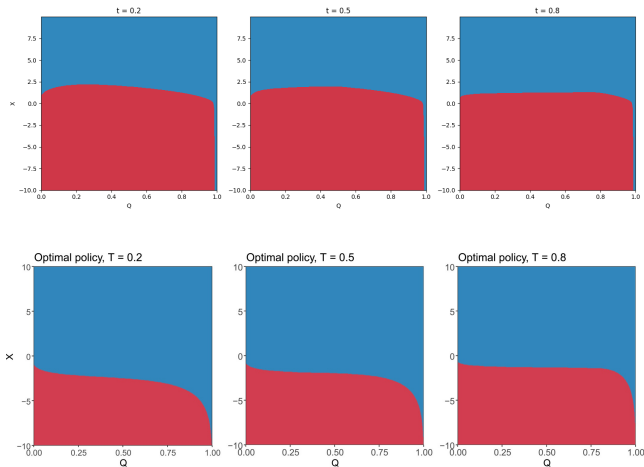


Figure: Robust (top) v. vanilla (bottom) policy under $\lambda = 10$, Gaussian prior $\gamma \sim N(0, 50^2)$, $\sigma = 5$. Arm pulled in blue region ($\pi = 1$).

- Results extend easily to a bandit with K different arms
 - Pull the arm with the highest index
- Threshold to pull is much more conservative under robustness
- Intuition: Exploration less useful due to rewards possibly shifting over time
 - Quadratic penalty captures this downside

- Formalize ambiguity/misspecification in statistical decision theory
- Link robust discrete time problems with continuous time counterparts
- Show that Neyman allocation remains optimal for best-arm identification & estimation